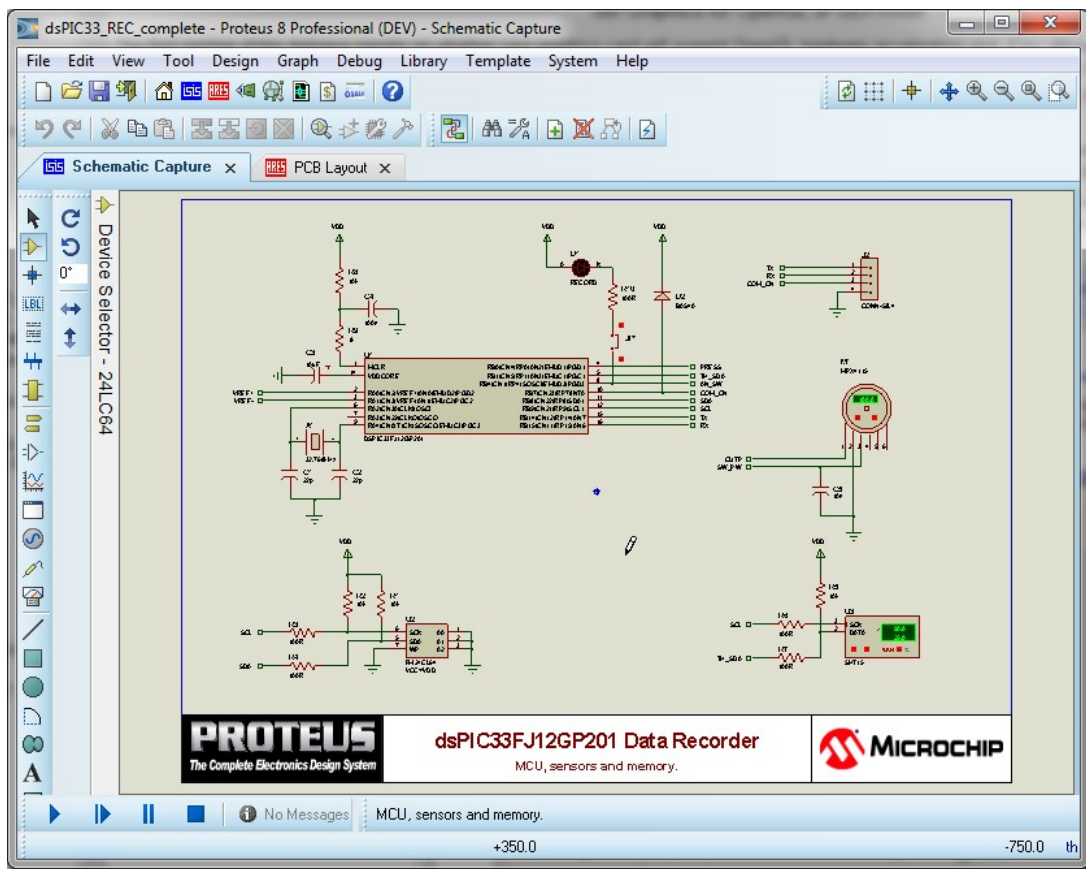


DESIGN OVERVIEW

Introduction

The circuit we are going to draw is shown below. This is reasonably straightforward schematic that will nonetheless allow us to cover most of the major features of the schematic capture package. Being a real-world design, it will also allow users to follow the design through the PCB phase of development via the PCB tutorial.



The dsPIC33 Data Recorder Sample Design

General Description

The dsPIC33 Data Recorder Tutorial circuit is a solid state recorder designed to collect three of the most important environmental variables:

- ☒ Atmospheric pressure
- ☒ Ambient temperature
- ☒ Relative humidity

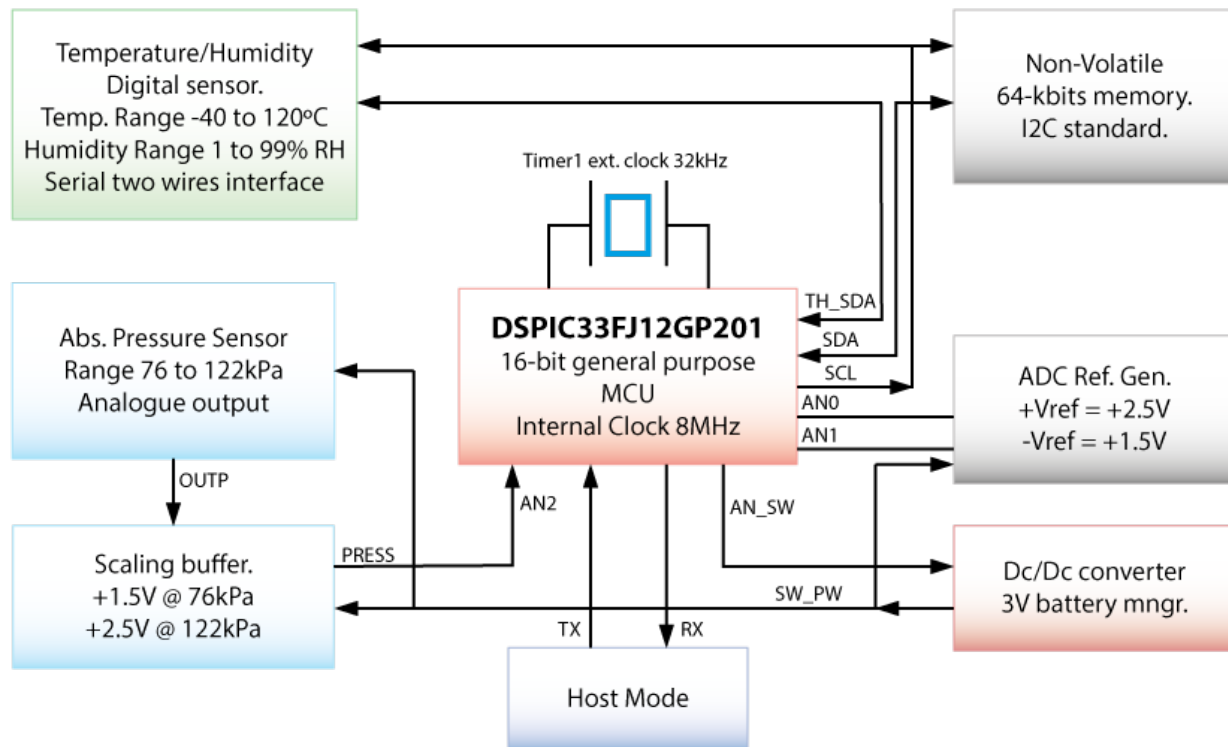
A number of low cost silicon transducers with built-in signal conditioning or digital interfaces have been used for measurement. This reduces the number of components to the minimum required for dsPIC33 interfacing.

All measurements get periodically stored into a non-volatile, low power memory in row binary format for later transfer to a host PC either by using a serial terminal or, more properly, a simple program (not supplied with the project). For the purpose of this project a serial terminal is used.

The circuit is designed to work with a battery for relatively long period in an unattended environment and, as such, a particular emphasis has been given to optimizing the power consumption and consequently the battery life. The recorder will sleep most of the time and will be woken up periodically only to accomplish the required recording operations.

Circuit Description

The following is a block diagram showing the behaviour and interaction of the circuit components.



Block diagram of the dsPIC Data Reorder Design

We'll cover the basics of the design construction as we progress through the tutorial but in order to keep the focus on the practical aspects of using the software, design decisions and theory will not be discussed in this document.